A Polished Weather Forecast

You are starting a job as an IT specialist in a weather station. To handle the large number of forecasts every day, the station uses the Polish logical notation[1] to store forecasts on their computer systems in a concise and unambiguous manner. To work with these systems, it is essential that you understand how the notation works.

In this notation, simple logical statements such as 'it will rain today' are assigned variables and are connected by logical operators, just like how variables and mathematical operators are used in maths to construct algebraic expressions and equations like *2p/q* or *4x + 5y = 7*.

Here, you are given a list of weather forecasts in English, followed by their 'translations' in the logical notation in **random order**. Variables are arbitrarily assigned to logical statements. Keep in mind that *or* is inclusive throughout this exercise (i.e. x *or* y means x *or* y *or both*).

1. It will rain today but not tomorrow.
2. If it rains today or there is both cold weather today and no snow tomorrow, it will be warm tomorrow.
3. If it rains today and tomorrow, it will be cold tomorrow but it will not snow.
4. It will not snow both today and tomorrow (but it might snow one of the days).
5. If it rains tomorrow, either it will not snow tomorrow or it will rain today.
6. It will be warm today or it will be cold and either snow or rain tomorrow.
7. There will be no snow today or tomorrow.
8. It is either warm today or it will neither rain today nor tomorrow.
9. If it rains or snows today, there will be cold weather tomorrow.
10. It will not be cold today.
11. It will be cold today and if it doesn't rain today, it will snow tomorrow.

(a) `AsNApw` (b) `AsKqAtw` (c) `CApuq` (d) `CApKvNtr` (e) `Nv` (f) `CKpwKqNt`
(g) `NKut` (h) `KvCNpt` (i) `KpNw` (j) `CwANtp` (k) `NAut`

Assignment 1:

Identify the correspondences between the forecasts (1-11) and their representations in logic notation (a-k).

[1] Polish notation (PN), also known as normal Polish notation (NPN), Łukasiewicz notation, Warsaw notation, Polish prefix notation or simply prefix notation, is a mathematical notation in which operators precede their operands. When Polish notation is used as a syntax for mathematical expressions by programming language interpreters, it is readily parsed into abstract syntax trees. Because of this, Lisp and related programming languages define their entire syntax in prefix notation.

Assignment 2:

Convert these statements in logic notation to weather forecasts in English:

a. `KNpq`

b. `CAuNrw`

c. `KNsCNpw`

Assignment 3:

Convert these weather forecasts into logic notation.

(a) Either it will be neither cold nor warm tomorrow, or it will rain tomorrow.

(b) It will snow today and if it is cold tomorrow, it will snow tomorrow.

(c) If it rains today but not tomorrow, it will not be cold either today or tomorrow (but it could be cold one of the days).

Assignment 4:

Explain briefly, in your own words, what the following symbols are used for in the logical notation:

(1) `A` (2) `C` (3) `K` (4) `N`

– Ishan Ganguly

Note: Since PLO 2021 Round 1 was held as an online contest, this problem was presented in an obfuscated form, with the real facts about the language modified/omitted.